



An Internship Report Of Web Development

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech (CSE)

By

Roll Number of Student:- 25SCS1003005314

Name of Student: Omji Singh

Batch: 2025-2029

CANDIDATE'S DECLARATION

I, **OMJI SINGH**, do hereby declare that the internship report titled "**Web Development**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:- _____

Student Name: Omji Singh

Roll No.: 25SCS1003005314

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank **Codec Technologies** for providing me the opportunity and a structured platform to learn and gain hands-on exposure in the domain of **Web Development**. I am also thankful to my institute, **IILM University, Greater Noida**, and the faculty of the **School of Computer Science and Engineering** for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: Omji Singh

Roll No.: 25SCS1003005314

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Internship Offer Letter



INTERNSHIP OFFER LETTER

Dear Omji Singh ,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Web Developer Intern
- Location & Mode: Pan India, Hybrid
- Duration : 25/07/2026 to 25/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

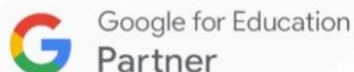
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: EI9E86-0116588288923
Date Of Issue - 25/07/2026




Dr. Anurag Shrivastava
Talent Acquisition Manager



INTERNSHIP COMPLETION CERTIFICATE



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the internship program undertaken in the domain of "Web Development", supported by Codec Technologies. The internship was completed as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to build a strong, industry-relevant foundation in modern web development — covering front-end technologies (HTML, CSS, Bootstrap, JavaScript) and responsive design principles, along with dynamic user interface creation and version control practices.

The internship followed a structured, module-wise curriculum that combined conceptual learning with hands-on practice across the design and functionality layers of the web development lifecycle. On successful completion of all the requirements, an Internship Completion Certificate was awarded by Codec Technologies

4.2 Organization Profile

Codec Technologies is an ISO-certified technology training and EdTech organization that runs structured, industry-aligned virtual internship and skill development programs for engineering and technology students across India. Codec Technologies partners with academic institutions to deliver domain-specific internship credentials — including Web Development — that combine self-paced learning modules, hands-on project assignments, and practical assessments designed to help students build employable, industry-ready technical skills.

4.3 Problem Statement

Modern businesses generate massive volumes of transactional and operational data across multiple data streams. However, raw data lacks business value unless properly aggregated, structured, and presented visually for decision-makers. Traditional static reports fail to provide immediate, actionable insight. The objective was to design and build a central **Data Analytics & Business Intelligence Dashboard** capable of processing data imports, computing key performance indicators (KPIs), dynamically

visualizing metrics through custom charts, and exporting actionable business summaries.

4.4 Project Objectives

- **Platform Architecture & Business Intelligence Foundations:** To conceptualize and construct a centralized Data Analytics & Business Intelligence Dashboard capable of transforming raw operational data into structured visual metrics for effective business decision-making.
- **Dynamic & Custom Dashboard Engineering:** To design and build flexible, component-based user interfaces in React.js that allow users to configure custom layout views, toggle modular analytical widgets, and adapt dashboards according to specific operational needs.
- **Real-Time KPI Monitoring Implementation:** To engineer dedicated tracking panels that continuously compute, aggregate, and render Key Performance Indicators (KPIs) to monitor real-time performance trends and critical business metrics.
- **Multi-Source Data Ingestion Pipelines:** To develop resilient input handling systems and backend routing pipelines capable of importing, parsing, and validating data payloads coming from diverse formats and external file sources.
- **Advanced Graphical Data Visualization:** To integrate interactive charting libraries (Chart.js / D3.js) for rendering visual analytical charts, including trend lines, distribution graphs, and comparative bar charts.
- **Automated & Exportable Report Generation:** To construct export utility modules that compile active dashboard state data into standardized, downloadable analytical reports for distribution and offline reporting.
- **Full-Stack Data Persistence & Integration:** To establish seamless RESTful API communication between the React frontend and Node.js backend while utilizing PostgreSQL database storage to ensure optimal schema performance.
- **Performance Optimization & State Management:** To implement efficient state management and optimized database querying techniques, ensuring fast dashboard rendering times and minimal response latency during large data imports.

- **User-Centric Responsive UI Design:** To build an intuitive, fully responsive dashboard across desktop and mobile devices.
- **Interactive Data Filtering & Aggregation:** To implement dynamic client-side filtering controls and date-range pickers that allow users to slice, dice, and aggregate large datasets instantaneously without full-page reloads.
- **Role-Based View Customization:** To structure modular component boundaries that support variable UI states based on user preferences and access levels for enterprise dashboard views.
- **REST API Endpoint Optimization:** To design clean, structured JSON API endpoints in Node.js/Express for efficient request payload handling, pagination, and fast response cycles.
- **System Scalability & Modular Architecture:** To apply strict modular software engineering practices, ensuring individual dashboard widgets, data models, and API services can be easily scaled or extended with minimal code refactoring.

4.5 Scope of the Project

The scope of this project encompasses the end-to-end full-stack development of a centralized Data Analytics & Business Intelligence Dashboard platform. It spans client-side dynamic UI rendering using React.js along with Chart.js and D3.js visualization engines to present multi-metric interactive charts, real-time KPI monitors, and customizable widget layouts. On the backend, the scope covers building Node.js and Express server layers to process multi-source data ingestion, compute aggregation metrics, and handle API payload routing. Additionally, it includes designing PostgreSQL relational database schemas for secure data persistence and establishing export utilities for generating downloadable analytical reports. The overall project was executed within a practical development environment provided by Codec Technologies, focusing on building a functional full-stack web application rather than live enterprise cloud deployment.

4.6 Technologies and Tools Used

- **Frontend Technologies:** HTML5, CSS3, JavaScript (ES6+), React.js, Chart.js / D3.js (for dynamic data visualization)
- **Backend Framework:** Node.js, Express.js

- **Database & Persistence:** PostgreSQL
- **API Architecture:** RESTful APIs, JSON
- **Development Tools & IDEs:** Visual Studio Code, Git, GitHub, Postman, Web Browser Developer Tools

4.7 System Architecture

The architecture of the Data Analytics & Business Intelligence Dashboard follows a multi-tiered client-server model designed to maintain clear separation of concerns, scalability, and fast execution across all layers of the application:

- **Presentation Layer (Frontend):** Built using React.js along with HTML5, CSS3, and JavaScript (ES6+). This layer handles user interaction, UI routing, component rendering, and client-side state management. It incorporates Chart.js and D3.js visualization engines to render interactive graphical charts, KPI panels, and custom dynamic dashboard layouts.
- **Application & Routing Layer (Backend):** Powered by Node.js and Express.js, serving as the core business logic engine. This layer manages RESTful API endpoints, processes incoming client requests, executes data ingestion pipelines, validates multi-source payloads, and performs numerical aggregation for analytical metrics.
- **Database & Persistence Layer:** Utilizes PostgreSQL as the primary relational database system. It securely stores structured operational data, user configuration preferences, dynamic dashboard layout states, and transactional analytics records for fast retrieval and relational querying.
- **Data Ingestion & Processing Pipeline:** Operates between the presentation layer and backend services to process multi-format file uploads and raw data streams, parsing and transforming raw inputs into structured formats compatible with database schemas and charting components.
- **Reporting & Analytics Export Engine:** Formulates compiled dashboard data and dynamic KPI metrics into exportable file formats, enabling users to generate downloadable analytical summaries directly from active browser sessions.

- **Development & Version Control Integration:** Relies on Git and GitHub for source code tracking and version control, alongside Visual Studio Code, Postman API testing suites, and browser developer tools for debugging and interface optimization.

4.8 Methodology

The 1-month Web Development internship at Codec Technologies was executed through a structured, week-by-week progressive methodology. The program combined theoretical learning with hands-on practice, focusing on building client-side interfaces, backend services, database structures, and dynamic visual dashboard components.

- **Week 1 (Modern Web Fundamentals & System Setup):** Focused on reviewing core ES6+ JavaScript concepts, asynchronous operations, dynamic DOM manipulation, HTML5 semantic structures, and environment setup using Visual Studio Code and Git for full-stack workflows.
- **Week 2 (Frontend Component Architecture & Data Visualization):** Centered on developing modular, dynamic UI views using React.js. Integrated Chart.js and D3.js libraries to build interactive graphs, KPI tracking components, and responsive dashboard layouts styled with modern CSS3 techniques.
- **Week 3 (Backend API Development & Database Integration):** Focused on constructing Node.js and Express server routing structures, building RESTful API endpoints, designing PostgreSQL relational database schemas, and setting up data ingestion handling for multi-source input payloads.
- **Week 4 (Dashboard Synthesis, Export Utilities & Evaluation):** Completed full-stack integration connecting the React frontend to backend services and database storage. Developed downloadable report export utilities, performed end-to-end debugging using Postman and Browser DevTools, and finalized project deliverables for evaluation and certification.

Week	Module	Description
Week 1	Modern Web Fundamentals & Setup	Reviewed core ES6 JavaScript, async operations, dynamic DOM setups, and environment configurations for full-stack workflows.
Week 2	Frontend Components & Data Visualization	Developed React.js dynamic views, created modular UI structures, and integrated Chart.js/D3.js for interactive graphical rendering.
Week 3	Backend Services & Database Integration	Configured Node.js REST API endpoints, designed database models, and established data import streams from multiple input formats.
Week 4	Dashboard Synthesis, Export & Evaluation	Implemented KPI tracking panels, added report export features, optimized database querying, and finalized project assessment for certification submission.

4.9 Expected Outcomes

- **Operational Business Intelligence Platform:** Successful development and deployment of a functional Data Analytics & Business Intelligence Dashboard capable of visualizing operational metrics and business performance indicators.
- **Modular Frontend Component Architecture:** Mastery in designing component-based user interfaces using React.js to create dynamic, reconfigurable dashboard widgets and intuitive layout views.

- • **Advanced Graphical Data Renderings:** Capability to integrate visualization libraries (Chart.js / D3.js) to convert complex datasets into clear, interactive charts, graphs, and trend indicators.
- • **Real-Time KPI Monitoring Capabilities:** Ability to calculate, aggregate, and display dynamic Key Performance Indicators (KPIs) for real-time tracking of operational metrics.
- • **Multi-Source Data Handling & Ingestion:** Technical skill in building resilient backend data ingestion routines in Node.js to receive, parse, and validate input payloads from diverse file formats.
- • **Relational Database Design & Persistent Storage:** Command over PostgreSQL database schema creation, table relationships, and efficient SQL querying to maintain data persistence across active client sessions.
- • **Exportable Business Summaries:** Implementation of structured report export utilities that enable users to convert active dashboard states into formatted, downloadable analytics files for offline review.
- • **Full-Stack REST API Integration:** Practical experience in building structured RESTful endpoints to connect client-side state handling seamlessly with server logic and persistent database engines.
- • **End-to-End Development Lifecycle Exposure:** Comprehensive understanding of the modern web software development life cycle, from UI design and API integration to debugging and project submission.

4.10 Certificates of Completion and Communication Proof

The official Internship Completion Certificate issued by Codec Technologies, validating the successful completion of the 1-month Web Development internship program, is attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

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